

QIAOYUE TANG

(+1)778-317-5073 | cathy.tangqiaoyue@gmail.com | qiaoyuet@cs.ubc.ca

EDUCATION

University of British Columbia

Ph.D. in Computer Science, under supervision of Mathias Lécuyer
Expected graduation: Late 2026 - Early 2027

Vancouver, BC

September 2021 – Now

University of British Columbia

MSc. in Statistics, GPA: 91/100, under supervision of Lang Wu & Aline Talhouk

Vancouver, BC

September 2017 – July 2019

University of British Columbia

BSc. in Statistics, Minor in Economics, GPA: 86/100

Vancouver, BC

September 2013 – June 2017

EXPERIENCE

Machine Learning Research Intern

Royal Bank of Canada, Borealis

May 2025 – October 2025

Vancouver, Canada

Machine Learning Engineer

Huawei Technologies, AI Application Research Center

August 2019 – August 2021

Shenzhen, China

Graduate Research Assistant

BC Cancer Research Center, OVCARE

May 2018 – June 2019

Vancouver, BC

Undergraduate Research Assistant

Vancouver General Hospital, Eye Care Center

September 2016 – January 2017

Vancouver, BC

RESEARCH

My research focuses on data privacy, including (1) designing efficient privacy-preserving algorithms with provable guarantees, (2) auditing privacy leakage in trained machine learning models, and (3) adapting existing tools to domain-specific problems. I have also worked on broader responsible AI topics, such as mitigating social bias and developing fairer models.

SELECTED PROJECTS

Private and Stable Test-time Adaptation

Project at UBC

- Incorporate differentially private training techniques to improve test-time adaptation (TTA) performances, where trained models adapt parameters in reducing error on new and distributionally different test data.
- Perform experiment and analysis for private TTA algorithms on pretrained ViT models and large-scale corrupted dataset ImageNet-C. [1]

Improving model fairness on vision and language downstream tasks

Project at RBC Borealis

- Propose a coherent framework that improves both prediction- and representation-level fairness when fine-tuning downstream tasks from pre-trained models with potential social biases.
- Apply the proposed framework to both vision and language datasets and pre-trained models and show improved fairness performance while maintaining downstream prediction accuracy. [2]

Privacy Auditing of Machine Learning Models without Retraining

Project at UBC

- Propose a framework using synthetic non-member data for evaluating membership inference attacks on the target model to audit its privacy level in practice.
- Perform theoretical analysis to provide rigorous guidelines for privacy auditing under the proposed framework. [4]
- Extending the framework in auditing large multi-model foundation models such as CLIP.

Correcting DP Bias in Adam's Second Moment Estimation

Project at UBC

- Analyze the interaction between DP and the Adam optimizer, and show that DP noise introduces bias in Adam’s second moment estimator which reduce DP-Adam to DP-SGD with momentum and violates the sign-descent hypothesis for Adam’s performance.
- Propose a variation of DP-Adam that corrects for the bias introduced by DP noise, and analyze its theoretical properties.
- Evaluate the proposed method which yields significant improvements (up to 3.5 percentage points of test accuracy) over DP-Adam. [5]
- Extend to investigate under heavy-tail class imbalance with applications to language model fine-tuning. [3]

Bug File Localization for Software Defect Tracking Systems

Project at Huawei

- Use historical data in closed issue documents to train a Sequence-to-Sequence model to capture implicit correspondence between user-submitted problem descriptions and the modified code file paths.
- Enhance model performance by incorporating source-code comment embedding and hard-coded rules on matching relevant files.
- Develop a utility tool based on the model to locate potential bug files in deep source-code directories to aid maintenance efficiency.
- Obtain > 55% hit-1 accuracy when recommending top-20 file paths when testing model performance in directories with > 10000 files. [6]

TECHNICAL SKILLS

Programming: Python, R, Bash

Developer Tools: Git, LaTeX, PyTorch, TensorFlow, JAX, Slurm

HONORS & AWARDS

Registration scholarship, Canadian AI 2026	April 2026
Funded Trainee Award, Advanced Machine Learning Training Network, CAN\$15000	December 2025
Student Travel Award, Theory and Practice of Differential Privacy	August 2024
Ph.D. Tuition Award, UBC	2021 - 2024
Outstanding Individual Award, Huawei, AARC	January 2021
Outstanding Team Award, Huawei, AARC	January 2021
MSc. International Tuition Award, UBC	September 2017, September 2018
Outstanding International Student Award, UBC	September 2013

SERVICE

Reviewer. ICML2026, ECCV 2026, CVPR 2026, AAAI2026, NIPS2025, AISTATS2024, NIPS2024

SELECTED PUBLICATIONS & PATENTS

* indicates equal contribution

- [1] Zefeng Li*, **Qiaoyue Tang***, Mathias Lécuyer, Evan Shelhamer. Private and Stable Test-time Adaptation with Differential Privacy. (In submission to ICML2026).
- [2] **Qiaoyue Tang**, Sepidehsadat Hosseini, Mengyao Zhai, Thibaut Durand, Greg Mori. FairNVT: Improving Fairness via Noise Injection in Vision Transformers. (In submission to TMLR).
- [3] **Qiaoyue Tang**, Alain Zhiyanov, and Mathias Lécuyer. On the Performance of Differentially Private Optimization with Heavy-Tail Class Imbalance. High-dimensional Learning Dynamics (workshop at ICML). 2025.
- [4] Mishaal Kazmi*, Hadrien Lautreite*, Alireza Akbari*, **Qiaoyue Tang***, Mauricio Soroco, Tao Wang, Sébastien Gambs, Mathias Lécuyer, et al. PANORAMIA: Privacy Auditing of Machine Learning Models without. The 38th Annual Conference on Neural Information Processing Systems (NeurIPS) 2024.
- [5] **Qiaoyue Tang**, Frederick Shpilevskiy, Mathias Lécuyer. DP-AdamBC: Your DP-Adam Is Actually DP-SGD (Unless You Apply Bias Correction) AAAI 2024 (**Oral**).
- [6] **Qiaoyue Tang**, Mathias Lécuyer. DP-SGD-LF: Improving Utility under Differentially Private Learning via Layer Freezing. Arxiv. 2022.
- [7] **Qiaoyue Tang**, Hailin Hu, Chenwei Liu, Wei Peng. A Bug Localization Method for Software Defect Tracking Systems using Sequence-to-Sequence Techniques. Patent, patent ID: 86883482CN01-203585
- [8] Chenwei Liu, **Qiaoyue Tang**. Triplet Convolutional Networks for Classifying Mixed-Type WBM Patterns with Noisy Labels. 2021 IEEE International Test Conference (ITC).
- [9] **Qiaoyue Tang**. Classification Strategies for High-dimensional Multiclass Imbalanced Data. 2019. *MSc. Thesis*